Conditional Latent-Prior Generative Compressed Sensing for Petrophysical Property Estimation from Sparse Core Measurements

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Abstract

Dense well logs provide continuous indirect observations of subsurface properties, whereas core measurements are sparse, expensive, and usually available only at selected depths. Direct supervised models can exploit log features, but they often use sparse target-well measurements only as additional inputs. Classical compressed sensing provides measurement consistency, but its fixed-basis sparsity assumption can be too rigid for cross-well petrophysical prediction. We propose Conditional Latent-Prior Compressed Sensing with Generative Models (CLP-CSGM), which learns a generative range of target-property windows and uses well logs to predict a latent prior. Sparse core measurements then refine the reconstruction through a test-time latent inverse problem. In low-data cross-well clay-volume prediction, CLP-CSGM Ridge improves over a direct autoencoder baseline in all tested training-size and measurement-ratio cells, reducing mean RMSE from 0.01905 to 0.01397. In a real-well F03-4 GR-only porosity stress test, CLP-CSGM Ridge attains the best average RMSE, with strongest gains in the sparsest measurement regimes. Ablations confirm that both the conditional prior and the sparse-measurement term are necessary, and that Ridge is more stable than an MLP latent prior in the tested low-data grids.

1 Introduction

Petrophysical property estimation is a central task in reservoir characterization. Wireline logs are dense along depth and can be acquired in most wells, but properties such as porosity, clay volume, or shale volume are usually calibrated by core or laboratory measurements that are sparse and expensive [Ellis and Singer, 2007, Asquith and Krygowski, 2004]. This mismatch creates a recurring inverse problem: a model must infer a continuous target-property profile from dense indirect logs while honoring a small number of direct measurements in the target well.

Machine-learning regressors are attractive in this setting because they can learn nonlinear relations between logs and target properties from previously interpreted wells [Bishop, 2006]. However, a direct map from logs to targets may underuse sparse core observations available in the target well. A common direct workaround is to concatenate the sparse measurement vector with the logs and train a supervised model from $[u, b]$ to y. Although this can be competitive, it treats the sparse measurements as features rather than as measurement evidence about the reconstructed signal.

Compressed sensing (CS) offers the complementary perspective: measurements should constrain the reconstruction through a forward model [Donoho, 2006, Candes et al., 2006a,b]. Classical CS usually assumes sparsity in a fixed basis, such as wavelets or discrete cosine atoms. In cross-well petrophysical prediction, this assumption can be too restrictive. Geological variability is not necessarily sparse in a universal basis, and fixed-basis residual correction may fail to outperform strong supervised baselines.

Compressed sensing with generative models (CSGM) replaces fixed-basis sparsity with a learned generative range [Bora et al., 2017]. Instead of seeking a sparse coefficient vector, one searches for a latent code z such that the generated signal $G(z)$ matches the measurements. This is well aligned with petrophysical windows: the decoder can represent plausible target profiles, while sparse core observations can select the profile that is consistent with the target-well evidence.

This paper proposes Conditional Latent-Prior CSGM (CLP-CSGM) for petrophysical property estimation from dense logs and sparse core measurements. The method trains an autoencoder decoder G on target-property windows and a ridge prior h that maps the dense log window u to an initial latent code $z_0 = h(u)$. At test time, the final latent code is obtained by solving a measurement-consistent inverse problem,

$$\hat{z} = \arg\min_{z} \|MG(z) - b\|_2^2 + \lambda\|z - z_0(u)\|_2^2, \qquad \hat{y} = G(\hat{z}). \tag{1}$$

The first term penalizes sparse core inconsistency; the second term keeps the solution close to the log-conditioned latent prior.

The contributions are:

- We formulate CLP-CSGM, a conditional generative compressed sensing method for petrophysical property windows.
- We provide a theoretical interpretation of the objective as a Tikhonov/MAP latent inverse problem and relate it to standard CSGM guarantees.
- We evaluate the method in low-data cross-well clay-volume prediction and in a real-well F03-4 GR-only porosity stress test.
- We show that CLP-CSGM Ridge improves over strong direct baselines in sparse-measurement regimes, and we use structural ablations to show that both the conditional latent prior and the sparse-measurement term are needed.

The remainder of the paper reviews related work, defines CLP-CSGM, describes the datasets and experimental protocol, reports quantitative and ablation results, and discusses the regimes in which the method is most appropriate.

2 Related work

Petrophysical prediction from logs. Well logs provide indirect but dense measurements of rock and fluid properties, and core-calibrated interpretation is a standard component of petrophysical analysis [Ellis and Singer, 2007, Asquith and Krygowski, 2004]. Data-driven models are now widely used to map logs to porosity, permeability, saturation, or shale-related properties [Ahmadi and Chen, 2019, Wood, 2020]. These models are effective when the training distribution is representative and labels are sufficient, but their usual prediction form is purely supervised. Sparse measurements in the target well are commonly used as additional features or for post-hoc calibration, not as explicit constraints in a forward measurement model.

Sparse compressed sensing. Classical compressed sensing reconstructs signals from incomplete linear measurements by assuming sparsity or compressibility in a known representation [Donoho, 2006, Candes et al., 2006a,b]. This view is appealing for sparse core assimilation because the measurement operator is known: a row of M selects the depths where the target was observed. Its

limitation is the structural prior. A fixed basis can be too restrictive for cross-well petrophysical windows, where changes in depositional setting, diagenesis, and log-response calibration may create variability that is not sparse in a universal transform.

Generative priors for inverse problems. CSGM replaces the sparse basis with the range of a trained generator [Bora et al., 2017]. The unknown signal is represented as $G(z)$, and reconstruction searches over z to minimize measurement mismatch. Bora et al. [2017] established recovery guarantees under set-restricted eigenvalue conditions and showed that generative priors can require fewer measurements than sparsity priors in image experiments. Subsequent work studied the power and limitations of generative compressed sensing [Kamath et al., 2020], invertible generative priors for inverse problems [Asim et al., 2020], and untrained architectural priors [Ulyanov et al., 2018]. These works support the central idea that an inverse problem can be solved by searching a structured generative representation rather than a fixed sparse basis.

Conditional inverse reconstruction. A useful distinction in the inverse-problem literature is between *direct conditional prediction* and *conditional inverse reconstruction*. In direct conditional prediction, auxiliary variables are used to predict the output directly, as in supervised regression, conditional generative modeling, or latent-code prediction followed by decoding [Mirza and Osindero, 2014, Sohn et al., 2015]. In conditional inverse reconstruction, by contrast, side information does not eliminate the inverse problem itself; rather, it regularizes the solution while measurement consistency remains explicit in the objective [Ongie et al., 2020, Lucas et al., 2018]. This distinction is central in our setting. Dense logs provide contextual information about the target interval, but sparse core-derived measurements remain the only direct observations of the property to be reconstructed. A method that simply predicts y from u, or even from $[u, b]$, does not preserve this separation of roles.

From this perspective, CLP-CSGM is neither a standard conditional generator nor a direct latent regressor. Standard conditional generative models use contextual information to generate samples directly from the conditioning variable [Mirza and Osindero, 2014, Sohn et al., 2015]. Such models are appropriate when the conditional input is intended to determine the output distribution by itself. Our problem is different: the dense logs u are informative but incomplete, while the sparse target-well measurements b carry direct evidence about the quantity of interest. For this reason, CLP-CSGM does not ask the model to synthesize y from u alone. Instead, it uses u only to define a target-specific latent anchor $z_0 = h(u)$, while the final estimate is obtained through a measurement-consistent latent inverse problem.

This formulation is also distinct from standard CSGM [Bora et al., 2017]. Classical CSGM searches the generator range by minimizing only the measurement mismatch. In CLP-CSGM, the latent search is additionally regularized toward a context-dependent prior predicted from dense logs. This difference is structural, not merely procedural: the term $\lambda\|z - z_0(u)\|_2^2$ remains active throughout the optimization and changes the reconstructed solution selected from the decoder range. Therefore, the proposed method should not be interpreted as "CSGM with a warm start." The logs do not only provide an initialization; they define a persistent latent regularizer that remains part of the inverse problem during test-time refinement.

A second relevant distinction is with respect to latent-regression baselines. One may first learn a latent representation and then regress directly from logs to latent codes, producing predictions of the form $G(h(u))$. This already introduces a learned target manifold, but it still performs *prediction without test-time assimilation*. Sparse target-well measurements do not alter the estimate once the latent code has been predicted. In CLP-CSGM, by contrast, the latent prior and the sparse

measurements play complementary roles: the prior $z_0 = h(u)$ defines a context-dependent center in latent space, while the measurement term $\|MG(z) - b\|_2^2$ pulls the estimate toward consistency with target-well observations. The ablation analysis in this paper confirms that neither component alone explains the observed gains; the strongest performance arises from their combination.

The proposed method also differs from fixed-basis sparse recovery. Classical compressed sensing assumes that the unknown signal is sparse or compressible in a predetermined transform basis [Donoho, 2006, Candes et al., 2006a,b]. That prior can be too rigid for petrophysical windows affected by cross-well distribution shift, where geological variability is not naturally aligned with a universal hand-crafted basis. CLP-CSGM replaces that assumption with a learned generative range, while preserving the core compressed-sensing principle that observed entries should appear explicitly in a data-fidelity term.

Position of this work. The proposed CLP-CSGM framework is positioned at the intersection of generative priors, conditional inverse reconstruction, and petrophysical estimation from sparse core measurements. Its contribution is not a new general theory of generative compressed sensing, but a task-specific formulation in which dense logs define a conditional latent prior, sparse target-well measurements define explicit test-time constraints, and reconstruction is restricted to a learned range of plausible property windows. This separation of roles distinguishes the method from direct regression, standard CSGM, and fixed-basis sparse recovery.

3 Method

3.1 Problem setup

Let $y \in \mathbb{R}^N$ denote a target-property window, such as clay volume or porosity, and let $u \in \mathbb{R}^p$ denote the corresponding dense log window. Sparse core measurements are modeled as

$$b = My + \eta, \tag{2}$$

where $M \in \mathbb{R}^{m \times N}$ is a known subsampling matrix, $m < N$, and η is measurement noise. The objective is to estimate y in a target well using both u and b.

Assumption 1 (Windowed generative structure)**.** *Target-property windows are well approximated by the range of a decoder $G : \mathbb{R}^k \to \mathbb{R}^N$, trained on target windows from training wells, with $k < N$.*

Assumption 2 (Log-conditioned latent predictability)**.** *The dense log window u contains enough information to predict a useful latent prior $z_0 = h(u)$, where h is fitted on encoded training windows.*

3.2 Generative prior

We train an autoencoder on standardized target windows. Its encoder maps y to a latent code $E(y)$, and its decoder maps a code back to a target window, $G(z)$. After training, only G is used in the inverse problem. This separates the signal prior from the sparse measurement operator: the decoder defines the admissible set of plausible property windows, whereas M specifies which depths are observed in the target well.

3.3 Conditional latent prior

The conditional prior is a ridge regression model

$$z_0 = h(u), \tag{3}$$

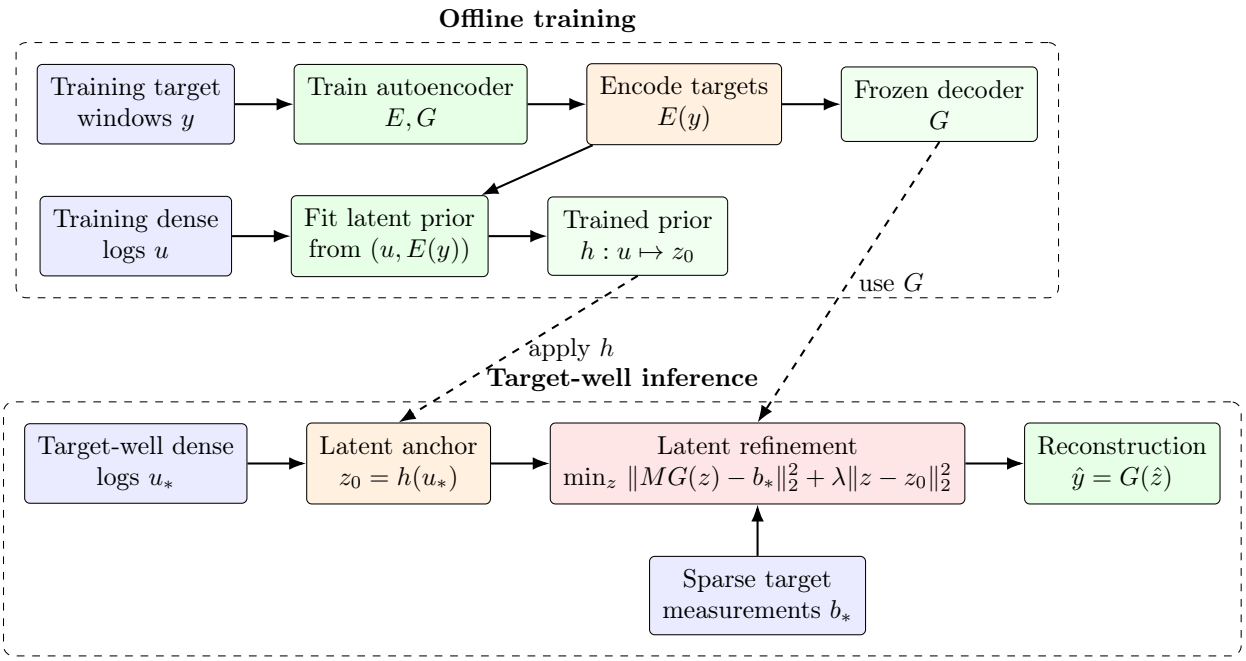

Figure 1: CLP-CSGM workflow. Offline, target windows train an autoencoder and dense logs are paired with encoded latents to fit the conditional prior h. For a target well, dense logs produce the latent anchor z_0, sparse measurements enforce data consistency, and latent optimization is decoded by G to obtain the final estimate.

trained to map log windows to encoded target latents. Ridge regression was chosen because it is stable in low-data regimes and regularizes correlated log features [Hoerl and Kennard, 1970]. The method is not tied to ridge regression: other predictors can be substituted for h, but the reported paper-facing variant is CLP-CSGM Ridge.

3.4 Test-time latent refinement

At validation and test time, the latent code is refined by

$$\hat{z} = \arg\min_z \underbrace{\|MG(z) - b\|_2^2}_{\text{sparse measurement consistency}} + \lambda \underbrace{\|z - z_0(u)\|_2^2}_{\text{conditional latent prior}}, \qquad \hat{y} = G(\hat{z}). \tag{4}$$

The regularization parameter λ is selected on validation windows for each measurement ratio and random seed.

Figure 1 summarizes the complete CLP-CSGM workflow and separates the operations that are learned offline from the operations performed for a target well. In the offline stage, target windows y train an autoencoder, whose decoder G is retained as the generative range used by the inverse problem. The same trained encoder maps target windows to latent codes $E(y)$; these latents, paired with dense log windows u, define the supervised training set for the latent prior $h : u \mapsto z_0$. In the target-well stage, the dense logs u_* are used only to compute the latent anchor $z_0 = h(u_*)$, while sparse observations b_* enter only through the data-fidelity term. The final estimate is therefore not a direct regression from $[u_*, b_*]$ to y_*; it is the decoded output of a latent code selected to remain close to the log-conditioned prior and consistent with the measured target entries.

Proposition 1 (MAP and Tikhonov interpretation)**.** *Assume* $b \mid z \sim \mathcal{N}(MG(z), \sigma_b^2 I)$ *and* $z \mid u \sim \mathcal{N}(z_0(u), \sigma_z^2 I)$*. Then the maximum a posteriori estimate of* z *minimizes Eq.* (4) *with* $\lambda = \sigma_b^2/\sigma_z^2$*, up to a positive multiplicative constant.*

Proof. The negative log posterior is, up to constants independent of z,

$$\frac{1}{2\sigma_b^2}\|MG(z) - b\|_2^2 + \frac{1}{2\sigma_z^2}\|z - z_0(u)\|_2^2.$$

Multiplying by $2\sigma_b^2$ gives Eq. (4). □

Proposition 2 (Relation to classical CSGM)**.** *If* $\lambda = 0$*, CLP-CSGM reduces to the standard CSGM measurement-consistency problem* $\min_z \|MG(z) - b\|_2^2$*. If* $\lambda \to \infty$*, the solution is forced toward the direct latent prediction* $z_0(u)$*, and the method approaches the decoder-only estimate* $G(z_0(u))$*.*

Proposition 3 (Existence of a minimizer)**.** *Assume that the decoder* $G : \mathbb{R}^k \to \mathbb{R}^n$ *is continuous and that* $\lambda > 0$*. For fixed* M*,* b*, and* z_0*, the objective*

$$\mathcal{J}(z) = \|MG(z) - b\|_2^2 + \lambda\|z - z_0\|_2^2$$

admits at least one global minimizer in $\mathbb{R}^k$*.*

Proof. Since G is continuous and M is linear, the map

$$z \mapsto \|MG(z) - b\|_2^2$$

is continuous. The quadratic regularization term

$$z \mapsto \lambda\|z - z_0\|_2^2$$

is also continuous and, because $\lambda > 0$, coercive in z. Therefore,

$$\mathcal{J}(z) \to +\infty \qquad \text{as} \qquad \|z\|_2 \to \infty.$$

Hence $\mathcal{J}$ is continuous and coercive on $\mathbb{R}^k$, so it attains its global minimum at some $\hat{z} \in \mathbb{R}^k$. □

Remark 1 (Role of the regularization parameter)*.* The parameter λ balances adherence to the log-conditioned latent anchor against sparse-measurement consistency. Small values of λ recover behavior closer to measurement-driven generative inversion, while large values increasingly restrict the solution toward the decoder-only conditional estimate $G(h(u))$. Thus, λ does not merely scale a penalty term; it determines how strongly dense contextual information is allowed to shape the final reconstruction relative to sparse direct observations. The validation grid implements this trade-off empirically.

Remark 2 (Representation error and decoder mismatch)*.* The proposed reconstruction is restricted to the image of the decoder G. Therefore, even when the sparse measurements are noiseless and the optimization is solved exactly, the reconstruction error cannot in general be smaller than the mismatch between the true signal and the decoder range. More precisely, if $y^\star$ denotes the unknown target signal and

$$\mathcal{M} = \mathrm{Im}(G)$$

denotes the learned manifold, then the method operates under an implicit approximation error of the form

$$\inf_{\tilde{y} \in \mathcal{M}} \|y^\star - \tilde{y}\|.$$

This term is not specific to the conditional latent prior; it is inherited from the use of a learned generative range itself. Consequently, CLP-CSGM should be interpreted as combining two sources of inductive bias: a manifold constraint induced by the decoder and a conditional latent regularization induced by the logs.

Remark 3 (Interpretation of the ablation study). The ablation study separates the contributions of three structural components: the learned manifold G, the conditional latent anchor $z_0 = h(u)$, and the sparse-measurement consistency term. The prior-only variant isolates the conditional latent prior without test-time assimilation, whereas the measurement-only variant isolates the generative inverse problem without dense contextual anchoring. The complete CLP-CSGM objective combines both mechanisms. Thus, improvements of the full method over both ablated variants empirically support the claim that its gains arise from the interaction between contextual latent regularization and sparse-measurement refinement, rather than from either component alone.

Remark 4 (Theoretical scope). CSGM theory provides recovery results when the measurement operator satisfies conditions such as S-REC over the generator range [Bora et al., 2017]. Our subsampling operators and learned autoencoder are practical choices for well-log windows, not a setting in which those guarantees are proven directly. The theory is used here to motivate the generative-prior formulation; empirical validation establishes performance for the studied petrophysical tasks.

3.5 Limitations

The method inherits representation error from the decoder: if a target profile is outside the learned range, no latent optimization can recover it exactly. The optimization problem is nonconvex, so restarts and validation are important. Finally, the method is computationally heavier than direct regression because it solves a latent inverse problem for each validation and test window.

4 Data and exploratory analysis

We use two related but distinct evaluation settings. The first is a cross-well clay-volume benchmark using noisy 30 dB logs, where the training wells are F02-1, F03-2, and F06-1 and the held-out test well is F03-4. The second is a single-well F03-4 porosity benchmark using only gamma ray as the dense input channel. Table 1 and Fig. 2 summarize the target distributions used by both settings.

Table 1: Dataset summary used to motivate the two experimental settings.

dataset	well	target	n_rows	depth_min	depth_max	target_mean	target_std	target_min	target_max
cross_train	F02-1	vc	165	1149.7056	1174.6992	0.1215	0.0407	0.0513	0.1934
cross_train	F03-2	vc	811	951.9000	1073.4000	0.1184	0.0490	0.0393	0.2688
cross_train	F06-1	vc	122	1133.1000	1151.2500	0.1223	0.0437	0.0427	0.2175
cross_test	F03-4	vc	832	946.5000	1071.1500	0.0824	0.0341	0.0112	0.1699
real_f03_gr	F03-4	porosity	1435	856.0500	1071.1500	0.2789	0.0142	0.2471	0.3264

The cross-well split is intentionally difficult. The held-out F03-4 Vc target has a lower mean than the training wells, indicating a distribution shift that direct supervised regressors must extrapolate across. This motivates a generative constraint on plausible target windows rather than relying only on a pointwise mapping from logs to targets. The low-data axis further increases this difficulty by increasing the sliding-window step and reducing the number of training windows.

Figure 3 shows the full Vc distributions for the cross-well experiment. The held-out F03-4 distribution is shifted toward lower Vc values relative to the training wells. This shift is important

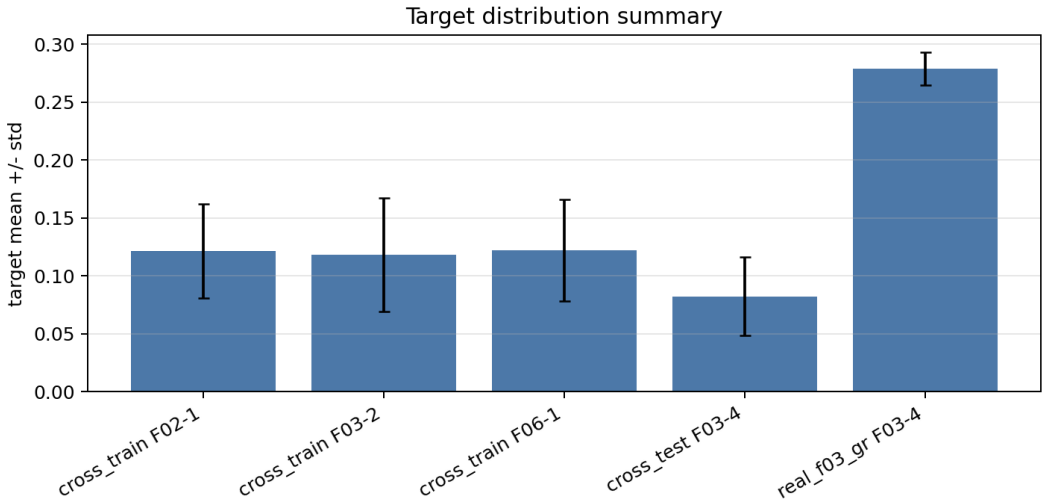

Figure 2: Summary of target distributions used in the CLP-CSGM experiments. The held-out cross-well Vc target has a lower mean than the training wells, while the F03 porosity case has a narrower target range.

for the article narrative: the method is not only interpolating within one well, but using sparse target-well measurements to adapt a learned latent prior under a visible distribution mismatch.

Table 2 reports simple Pearson correlations between input channels and targets. Gamma ray is strongly correlated with Vc in both cross-well train and test splits, whereas the real F03 porosity experiment uses gamma ray only despite a weak linear correlation with porosity. This GR-only choice is deliberate: it weakens the dense-input predictor and tests whether sparse target-well measurements can improve a log-conditioned generative prior.

Table 2: Channel-target Pearson correlations. These values are descriptive only and are not used as model inputs.

dataset	channel	target	pearson_r
cross_train	sonic	vc	-0.1642
cross_train	rhob	vc	0.0807
cross_train	gr	vc	0.9733
cross_train	ai	vc	0.1672
cross_train	vp	vc	0.1819
cross_test	sonic	vc	-0.1606
cross_test	rhob	vc	-0.0061
cross_test	gr	vc	0.9772
cross_test	ai	vc	0.1330
cross_test	vp	vc	0.1077
real_f03	ac	porosity	0.9996
real_f03	gr	porosity	-0.1759

Figure 4 visualizes the same channel-target correlations. In the cross-well Vc task, GR is the dominant linear correlate in both train and held-out wells, which explains why direct log-based baselines are not trivial competitors. In the F03 porosity task, however, GR is weakly and negatively correlated with porosity, whereas AC is almost perfectly correlated. Using GR only therefore creates

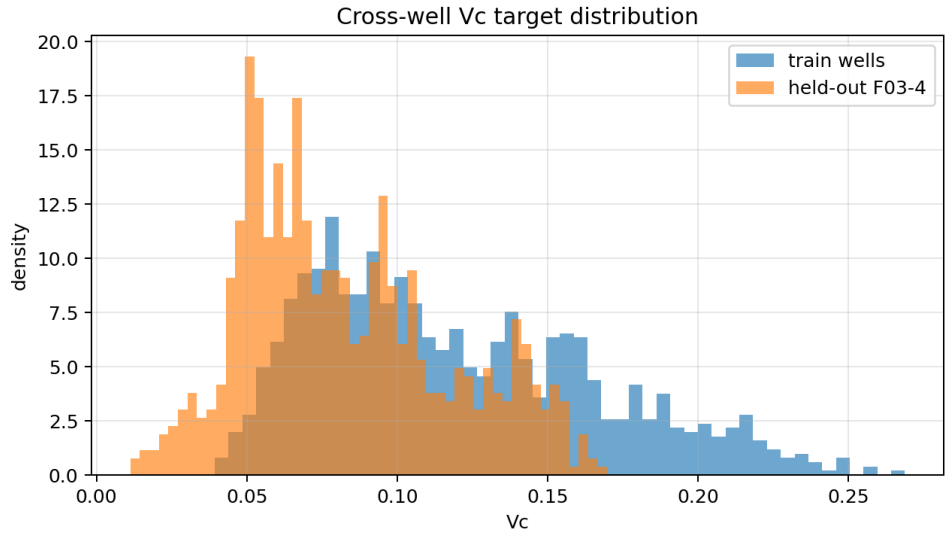

Figure 3: Cross-well Vc target distributions for training wells and held-out F03-4. The held-out test distribution is shifted toward lower Vc, motivating a measurement-consistent method rather than a purely supervised predictor.

a deliberately harder dense-input setting where the sparse measurements and the latent prior have a clear role.

Figure 5 complements the correlation table by showing the real F03-4 GR and porosity profiles along depth. The GR-only experiment is therefore a deliberately weakened dense-input setting: porosity varies smoothly over the interval, while GR alone is not a near-deterministic porosity proxy. This makes the sparse porosity observations central to the reconstruction task.

The EDA therefore supports the experimental design. Cross-well Vc evaluates low-data transfer under sparse target measurements, while the F03 GR-only case tests whether CLP-CSGM can recover useful porosity structure when the dense input alone is intentionally limited.

5 Experimental setup

5.1 Cross-well Vc low-data benchmark

The cross-well benchmark predicts Vc windows of length $N = 64$. Training and validation windows are built from F02-1, F03-2, and F06-1; F03-4 is held out for testing. The dense input u concatenates sonic, density, gamma ray, acoustic impedance, and V_p windows. Sparse observations b are generated by a coordinate subsampling matrix M with measurement ratios $\rho \in \{0.05, 0.10, 0.20\}$. The low-data regime is controlled by the sliding window step: step values $\{8, 16, 32\}$ yield approximately $\{104, 52, 27\}$ training windows.

For each step and measurement ratio, results are averaged over seeds $\{7, 23, 41\}$. The CLP-CSGM decoder is trained on standardized training targets, the ridge prior maps log windows to encoded latents, and λ is selected by validation RMSE.

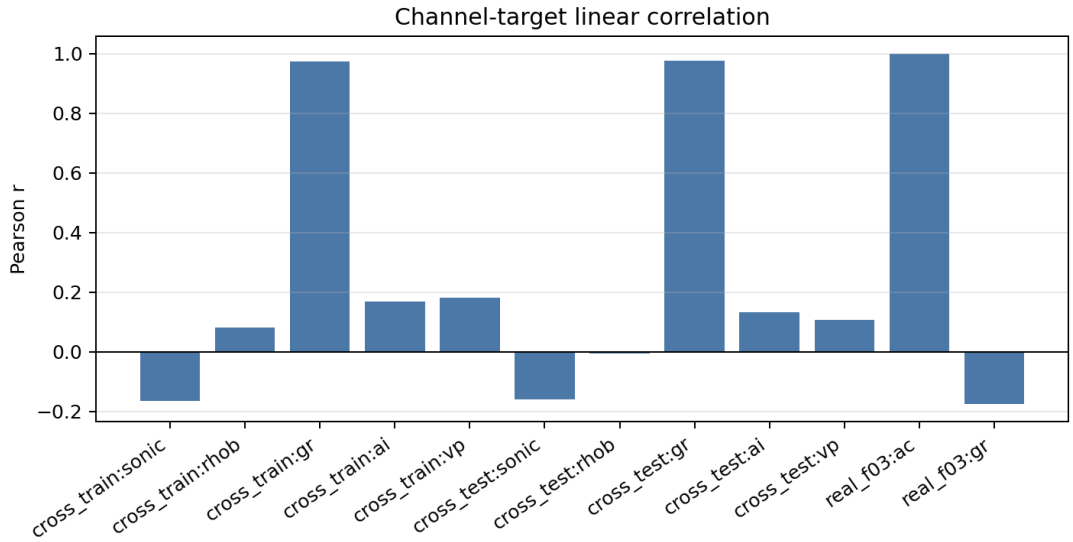

Figure 4: Channel-target Pearson correlations for the cross-well Vc and F03 porosity data. GR is highly informative for Vc but weak for F03 porosity, motivating the GR-only porosity benchmark as a difficult target-well test.

5.2 Real-well F03 GR-only porosity benchmark

The real-well experiment uses F03-4 AC, GR, and porosity data, but the dense input is intentionally restricted to GR only. Windows have length $N = 64$ and are split contiguously by depth: 823 training windows, 274 validation windows, and 275 test windows. Sparse porosity measurements are simulated by coordinate subsampling with $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$. This experiment should be interpreted as a stress test of sparse-measurement assimilation under weak dense context, not as a claim that GR-only is the preferred operational petrophysical configuration.

5.3 Implementation details

All target windows are standardized using the training-set mean and standard deviation before autoencoder training and latent optimization; predictions are mapped back to the original target scale before metric computation. The CLP-CSGM generator is a multilayer-perceptron autoencoder with one hidden layer of width 128 in both encoder and decoder, a latent dimension $k = 16$, ReLU activations, and a linear output layer. The autoencoder is trained for 200 epochs with Adam, learning rate 10^{-3}, weight decay 10^{-5}, and batch size 64.

After autoencoder training, the encoder maps each standardized training target window to a latent code. The paper-facing prior $h(u)$ is a Ridge regressor with $\alpha = 1.0$, fitted from standardized log windows to standardized latent codes. For the prior-class sensitivity experiment, the MLP prior uses two hidden layers of width 128, L_2 penalty 10^{-4}, learning rate 10^{-3}, a maximum of 500 iterations, and early stopping.

At validation and test time, latent refinement is optimized with Adam for 400 iterations, learning rate 0.05, and three restarts. The first restart is initialized at $z_0 = h(u)$; the remaining restarts add small Gaussian perturbations. The regularization parameter is selected on the validation split from

$$\lambda \in \{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}\}.$$

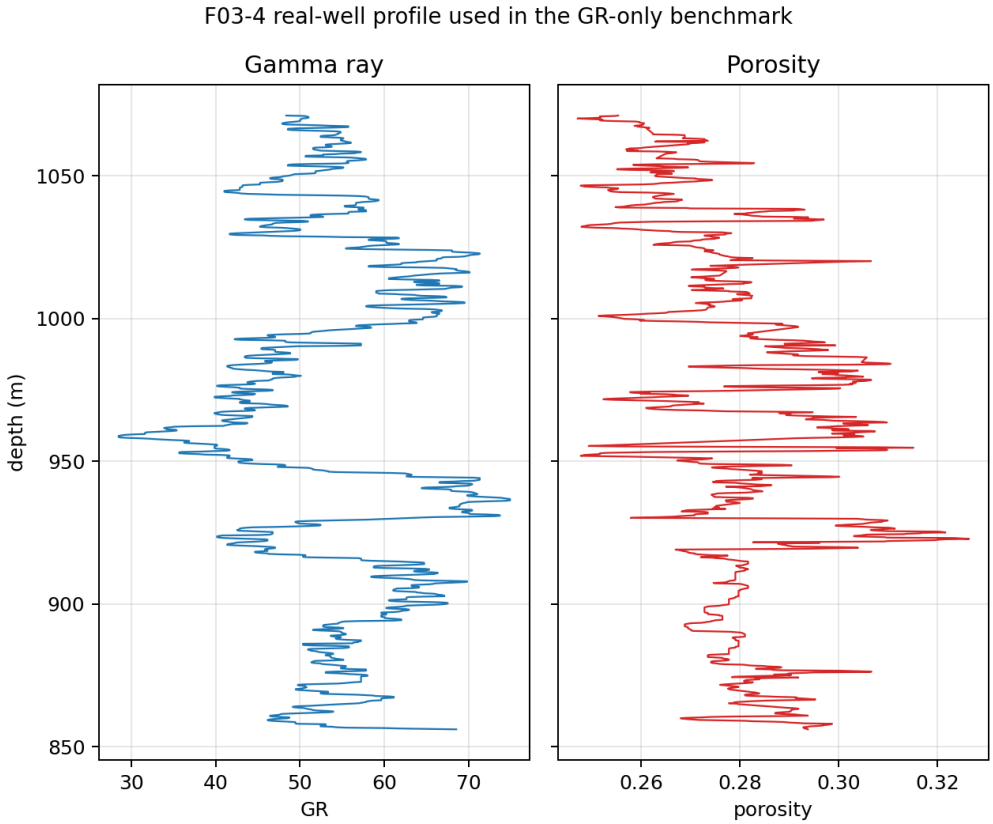

Figure 5: Real F03-4 depth profiles used in the GR-only porosity benchmark. Gamma ray is the only dense input, while porosity is the target profile to be reconstructed from sparse observations and the conditional latent prior.

For the measurement-only ablation, $\lambda = 0$ and the latent search is initialized from zero rather than from $h(u)$. For the prior-only ablation, no latent optimization is performed and the prediction is $G(h(u))$.

5.4 Baselines and metrics

The main baselines are:

- ML only: supervised MLP using dense logs only.
- MLP $[u, b]$: direct MLP using concatenated logs and sparse measurements.
- PCA $[u, b]$: PCA target representation with an MLP from $[u, b]$ to PCA scores.
- AE $[u, b]$: autoencoder target representation with an MLP from $[u, b]$ to latent codes.

The proposed method is reported as CLP-CSGM Ridge. We focus on RMSE, MAE, and relative ℓ_2 error. The previous LFISTA/SIR-CS line is not included in the main comparison because it is a different proposed method with a fixed-basis sparse residual prior rather than a conditional generative prior.

6 Results

6.1 Cross-well low-data results

Table 3 compares CLP-CSGM Ridge against the strongest direct baseline, AE $[u, b]$, for every low-data and sparse measurement cell. CLP-CSGM Ridge wins in all nine cells. The relative RMSE gap is modest at step 8 and $\rho = 0.05$, but becomes large as the low-data regime becomes more severe, reaching improvements above 38% for step 32 at $\rho = 0.10$ and $\rho = 0.20$. Figure 6 visualizes the same pairwise comparison and shows that the advantage is not confined to a single measurement ratio: within each training-size panel, the CLP-CSGM curve remains below the AE curve across the sparse-measurement grid.

Table 3: Cross-well Vc comparison between CLP-CSGM Ridge and AE $[u, b]$. Negative gap means CLP-CSGM has lower RMSE.

step	n_train_approx	rho	clp_csgm_rmse	ae_ub_rmse	gap_vs_ae_pct
8	104	0.05000	0.01541	0.01709	-9.84267
8	104	0.10000	0.01432	0.01635	-12.40651
8	104	0.20000	0.01167	0.01633	-28.52507
16	52	0.05000	0.01493	0.01891	-21.01886
16	52	0.10000	0.01391	0.01803	-22.81414
16	52	0.20000	0.01247	0.01808	-31.00344
32	27	0.05000	0.01562	0.02132	-26.71650
32	27	0.10000	0.01484	0.02501	-40.67353
32	27	0.20000	0.01259	0.02035	-38.15454

Figure 7 places this pairwise comparison in the full benchmark context. The direct MLP and PCA baselines become unstable in the low-data cells, with errors far above the ML-only and AE baselines. This behavior is informative but not the main evidence for the proposed method, because these direct models are clearly fragile in this regime. The main competitor is AE $[u, b]$, which uses the

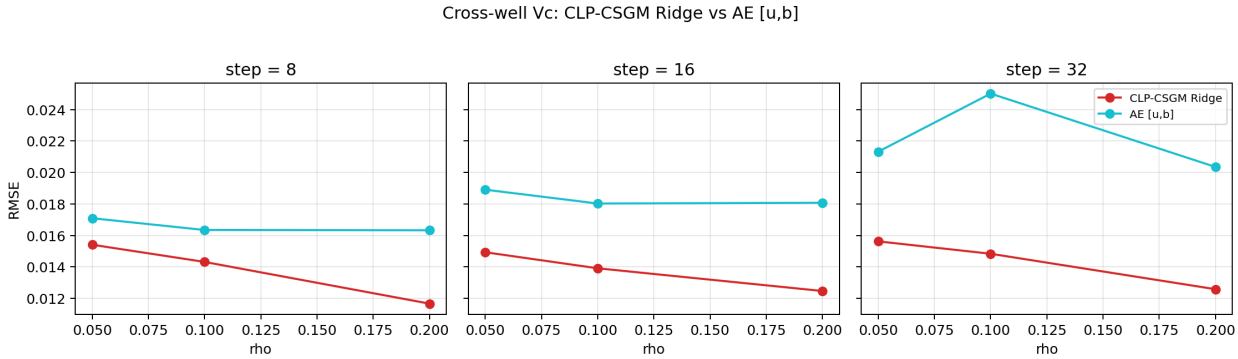

Figure 6: Cross-well Vc RMSE for CLP-CSGM Ridge and AE $[u, b]$ across the low-data grid. Each panel fixes the sliding-window step and varies the sparse measurement ratio.

same autoencoder representation family as a direct supervised latent regressor. CLP-CSGM Ridge consistently stays below AE $[u, b]$ across the measurement ratios, so the gain is not an artifact of comparing only against weak baselines; it is visible against the strongest direct representation-learning baseline.

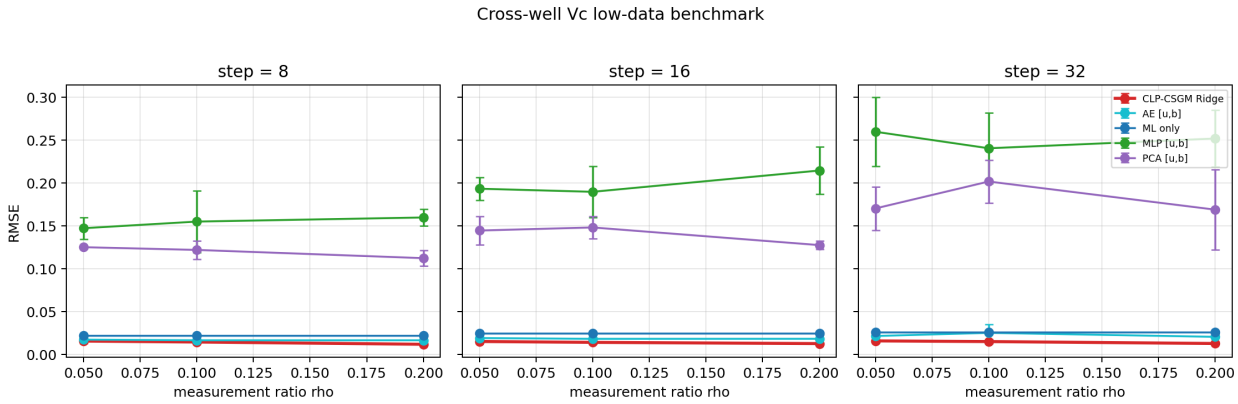

Figure 7: Cross-well Vc RMSE for all main benchmarks. CLP-CSGM Ridge remains the best method across the low-data grid, while direct MLP and PCA baselines degrade sharply when training windows are scarce.

The overall ranking in Table 4 shows the same pattern. Across the nine step–ρ cells, CLP-CSGM Ridge obtains mean RMSE 0.01397, compared with 0.01905 for AE $[u, b]$. This corresponds to a relative gap of about 26.65% against the strongest direct baseline.

6.2 Real-well F03 GR-only results

Table 5 reports the F03-4 GR-only porosity benchmark. CLP-CSGM Ridge has the best mean RMSE across measurement ratios, 0.01360 versus 0.01398 for AE $[u, b]$. The gain is concentrated in the sparse-measurement regime: CLP-CSGM Ridge is clearly better at $\rho = 0.20$ and $\rho = 0.30$. At higher measurement ratios, AE $[u, b]$ becomes slightly better. Figure 8 makes this crossover explicit: the proposed method is most useful when target-well measurements are scarce, whereas the direct AE baseline benefits once more entries of b are available as input features.

Table 4: Overall cross-well Vc ranking across all low-data cells.

method	mean_rmse	mean_relative_l2	cells
CLP-CSGM Ridge	0.01397	0.18131	9
AE [u,b]	0.01905	0.25464	9
ML only	0.02403	0.32932	9
PCA [u,b]	0.14654	2.08220	9
MLP [u,b]	0.20112	2.79301	9

Table 5: Real-well F03-4 GR-only porosity benchmark by measurement ratio.

rho	method	rmse	mae	relative_l2
0.20000	CLP-CSGM Ridge	0.01575	0.01365	0.05909
0.20000	AE [u,b]	0.01797	0.01560	0.06737
0.20000	PCA [u,b]	0.02079	0.01712	0.07791
0.20000	ML only	0.02546	0.02219	0.09550
0.20000	MLP [u,b]	0.03189	0.02731	0.11949
0.30000	CLP-CSGM Ridge	0.01449	0.01245	0.05437
0.30000	AE [u,b]	0.01536	0.01330	0.05760
0.30000	PCA [u,b]	0.02241	0.01834	0.08398
0.30000	ML only	0.02546	0.02219	0.09550
0.30000	MLP [u,b]	0.03184	0.02716	0.11924
0.40000	AE [u,b]	0.01296	0.01107	0.04856
0.40000	CLP-CSGM Ridge	0.01342	0.01141	0.05034
0.40000	PCA [u,b]	0.02206	0.01790	0.08263
0.40000	ML only	0.02546	0.02219	0.09550
0.40000	MLP [u,b]	0.03728	0.03245	0.13966
0.50000	AE [u,b]	0.01234	0.01065	0.04625
0.50000	CLP-CSGM Ridge	0.01246	0.01061	0.04673
0.50000	PCA [u,b]	0.02358	0.01909	0.08834
0.50000	ML only	0.02546	0.02219	0.09550
0.50000	MLP [u,b]	0.04466	0.03965	0.16717
0.60000	AE [u,b]	0.01126	0.00955	0.04218
0.60000	CLP-CSGM Ridge	0.01186	0.01006	0.04447
0.60000	PCA [u,b]	0.02417	0.01960	0.09056
0.60000	ML only	0.02546	0.02219	0.09550
0.60000	MLP [u,b]	0.04423	0.03873	0.16568

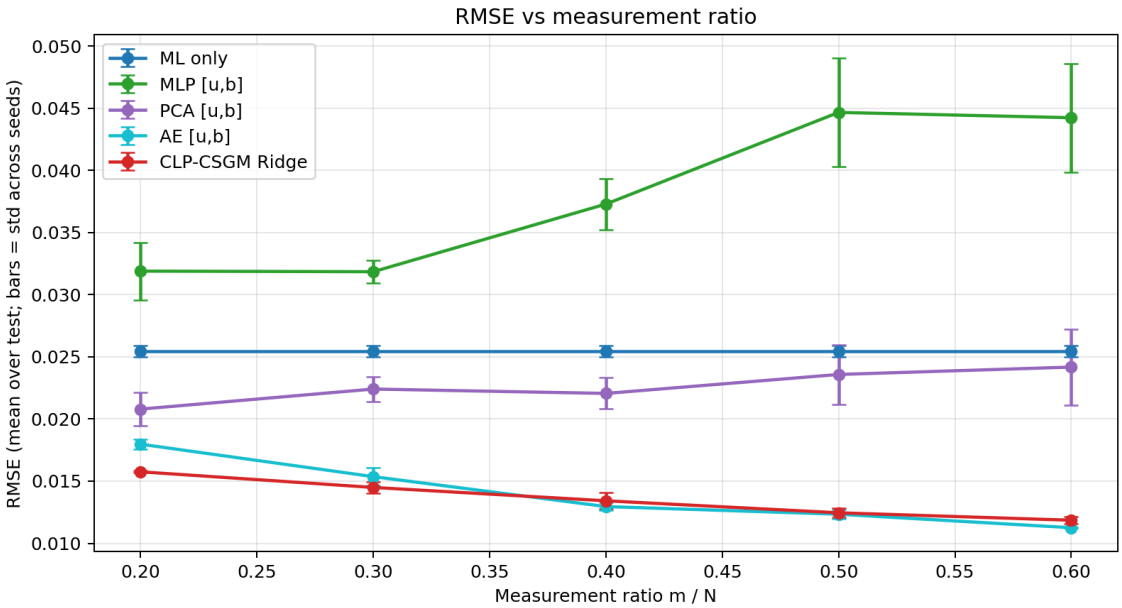

Figure 8: Real-well F03-4 GR-only porosity RMSE by measurement ratio. CLP-CSGM Ridge is strongest at the sparsest ratios, while AE $[u, b]$ becomes competitive as more target measurements are available.

The MAE and relative-ℓ_2 curves in Figs. 9–10 confirm that the RMSE behavior is not metric-specific. CLP-CSGM Ridge stays close to or below AE $[u, b]$ across the sparse-to-moderate ratios, while the direct MLP baseline deteriorates as more sparse-measurement features are added. This suggests that naively concatenating b to u can inject high-variance features, whereas the proposed method uses b only through the physical measurement residual.

Figure 11 expresses the same result as RMSE gain over the ML-only baseline. Positive values indicate improvement over using GR alone. CLP-CSGM Ridge and AE $[u, b]$ are the only methods with large positive gains across all ratios, but CLP-CSGM Ridge has the stronger gain when measurements are sparse. The direct MLP $[u, b]$ curve is negative, reinforcing that sparse measurements should be incorporated through a measurement model rather than only as raw input features.

The qualitative diagnostics show how these aggregate gains appear in reconstructed profiles. Figure 12 shows representative test windows: MLP $[u, b]$ can produce oscillatory outputs, while PCA, AE, and CLP-CSGM Ridge are smoother. Figure 13 evaluates pooled pointwise agreement and shows the highest displayed correlation for CLP-CSGM Ridge, indicating better ordering of low and high porosity values. Finally, Fig. 14 reconstructs the test interval along depth and shows that CLP-CSGM tracks the observed porosity trend without the high-frequency instability of the direct MLP baseline.

This behavior is consistent with the design of CLP-CSGM. When measurements are very sparse, the log-conditioned latent prior prevents the inverse problem from fitting arbitrary sparse observations, while the measurement-consistency term corrects the weak GR-only prior. When measurements become denser, a direct AE baseline can use the $[u, b]$ feature vector effectively and the extra latent optimization may offer less advantage.

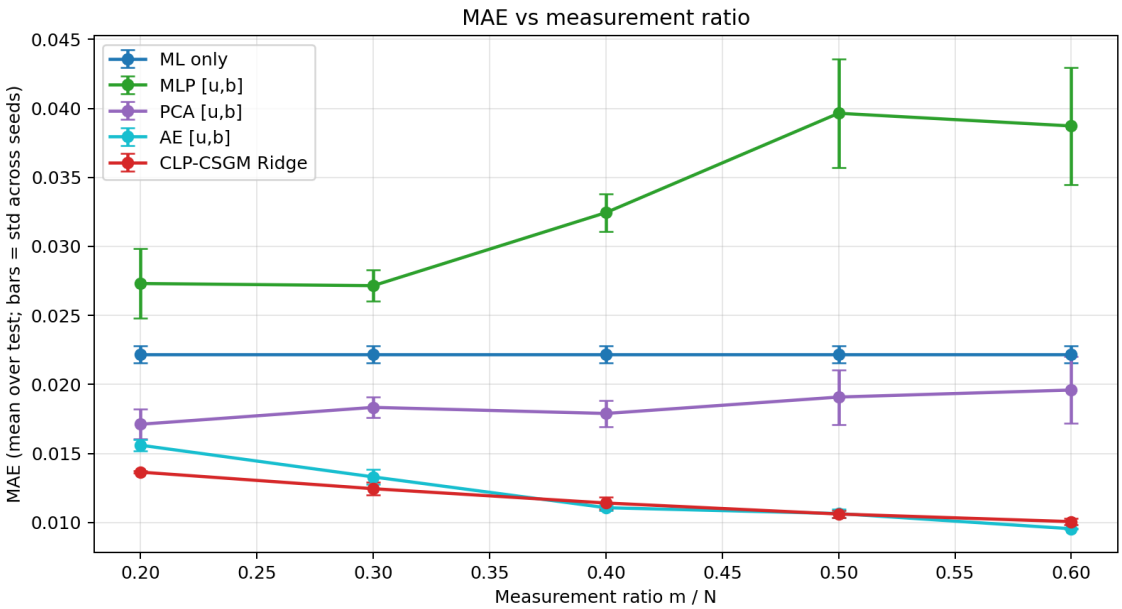

Figure 9: Real-well F03-4 GR-only MAE by measurement ratio. The ranking is consistent with RMSE: CLP-CSGM Ridge is strongest in the sparse regime and remains competitive as the measurement ratio increases.

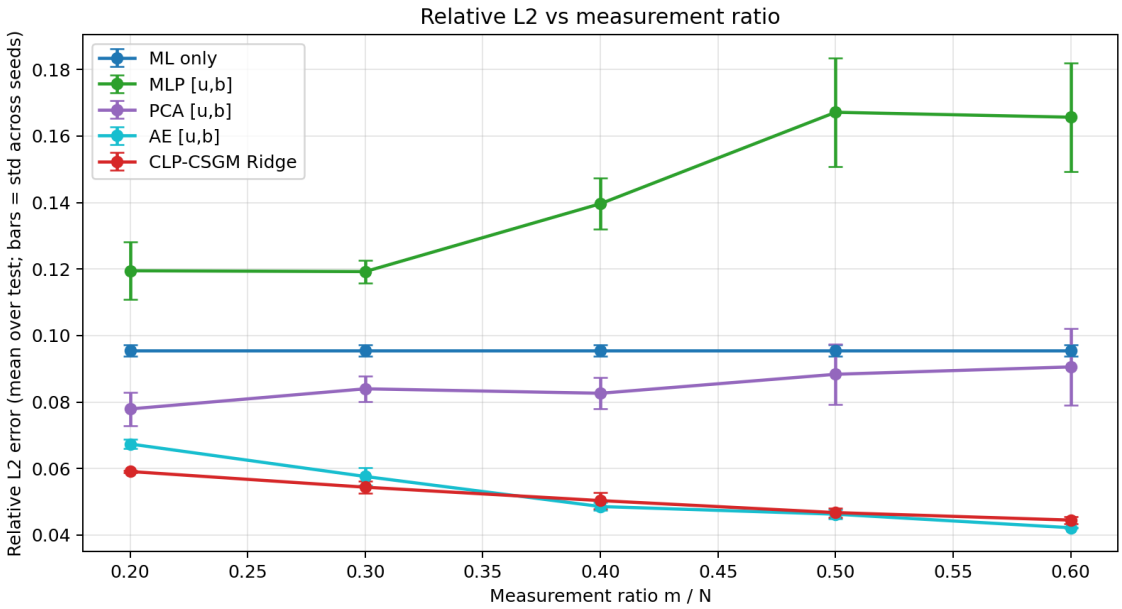

Figure 10: Relative ℓ_2 error on the real-well F03-4 GR-only benchmark. The relative-error view preserves the same conclusion as RMSE and MAE, indicating that the gain is not caused by scale effects alone.

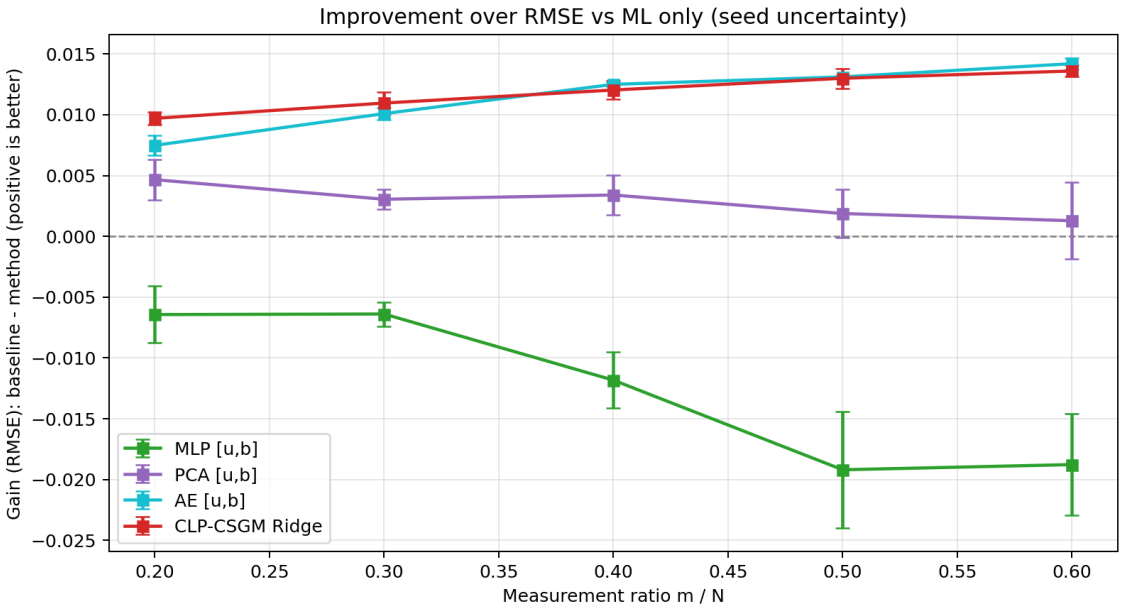

Figure 11: RMSE gain over the ML-only baseline on F03-4. CLP-CSGM Ridge gives the largest gain at low measurement ratios and remains consistently positive across the full range.

6.3 Ablation analysis

The ablation study isolates the two structural components of CLP-CSGM Ridge: the conditional latent prior and the sparse-measurement refinement. The prior-only variant uses $\hat{y} = G(h(u))$, without b. The measurement-only variant solves the CSGM objective with $\lambda = 0$, removing the log-conditioned anchor. Table 6 and Fig. 15 show that the full method improves over both ablations in every cross-well low-data cell. The gain over prior-only ranges from 6.4% to 24.8%, while the gain over measurement-only ranges from 39.4% to 61.3%. This confirms that the cross-well result is not explained solely by the decoder or solely by sparse measurements; the advantage comes from their combination.

Table 6: Cross-well Vc ablation results. Gains are RMSE reductions of CLP-CSGM Ridge relative to the corresponding ablation.

Step	ρ	CLP-CSGM	Prior-only	Meas.-only	AE $[u, b]$	Gain vs prior (%)	Gain vs meas. (%)
8	0.0500	0.0145	0.0160	0.0368	0.0171	9.7293	60.6685
8	0.1000	0.0141	0.0160	0.0297	0.0164	12.2869	52.6223
8	0.2000	0.0121	0.0160	0.0251	0.0164	24.8219	51.8636
16	0.0500	0.0140	0.0150	0.0362	0.0189	6.4056	61.2715
16	0.1000	0.0138	0.0150	0.0295	0.0180	7.9251	53.2853
16	0.2000	0.0129	0.0150	0.0213	0.0182	14.0183	39.4268
32	0.0500	0.0146	0.0157	0.0341	0.0213	7.1571	57.1221
32	0.1000	0.0147	0.0157	0.0271	0.0283	6.6852	45.7000
32	0.2000	0.0131	0.0157	0.0219	0.0205	17.1104	40.2930

The real-well F03 GR-only ablation in Table 7 and Fig. 16 shows the same decomposition under a weak dense context. CLP-CSGM Ridge improves over prior-only by 18.9% at $\rho = 0.20$ and by 38.9% at $\rho = 0.60$, showing that sparse porosity observations add information beyond $G(h(u))$.

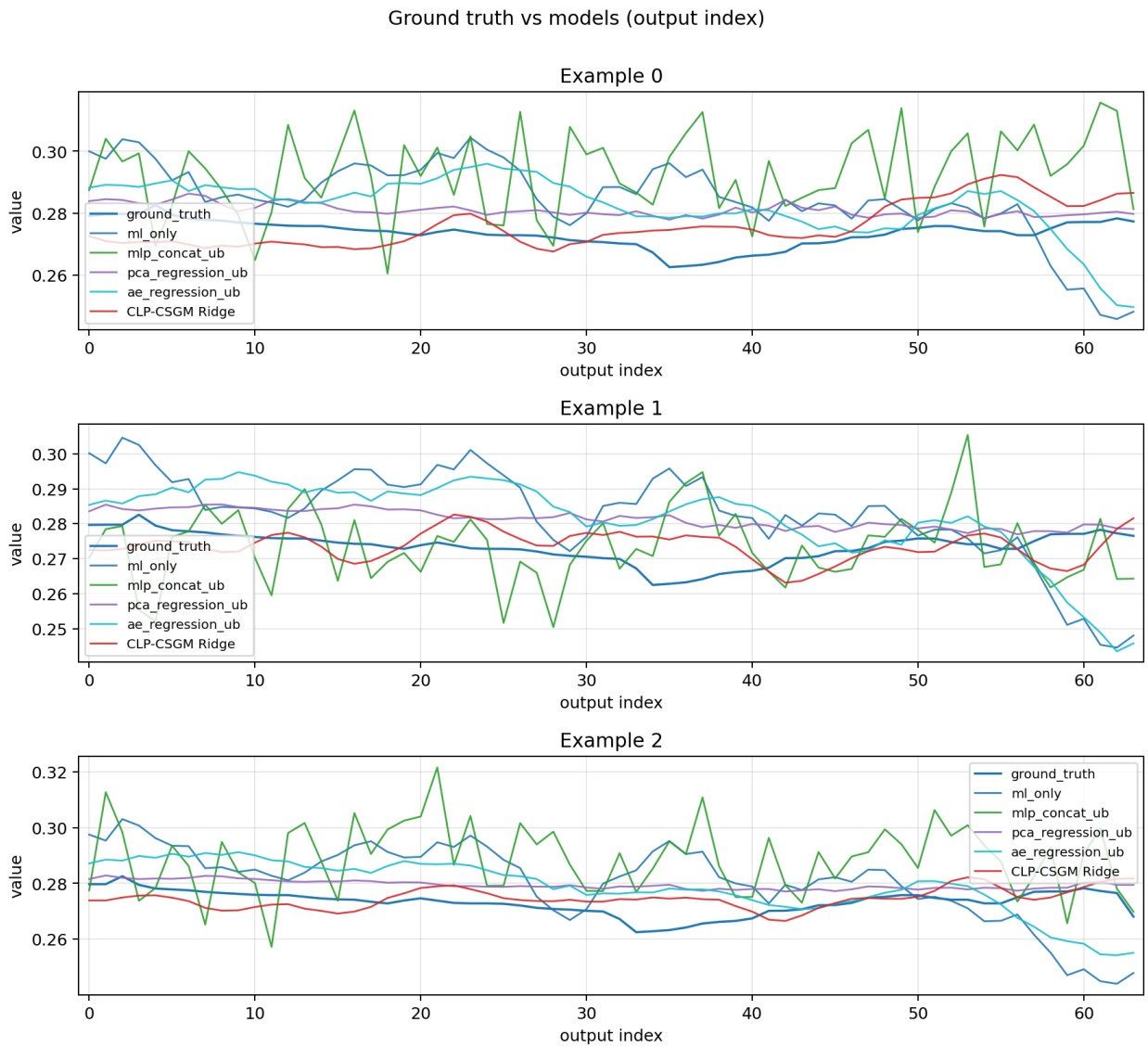

Figure 12: Example F03-4 test windows. CLP-CSGM Ridge produces smooth reconstructions that follow the ground-truth trend, while direct MLP $[u, b]$ is visibly more oscillatory.

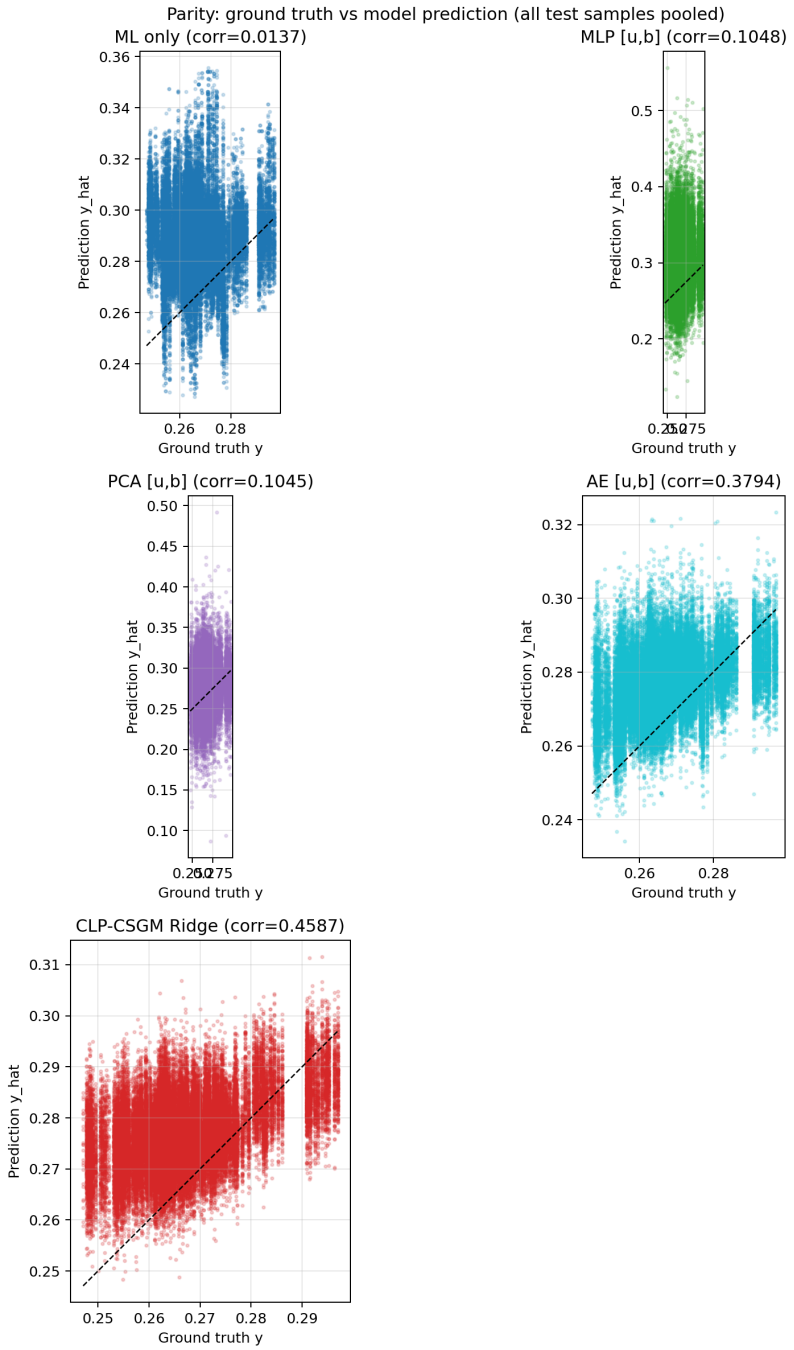

Figure 13: Pooled parity plots for F03-4 test windows. CLP-CSGM Ridge has the highest displayed correlation, indicating better ordering of high and low porosity values across the test block.

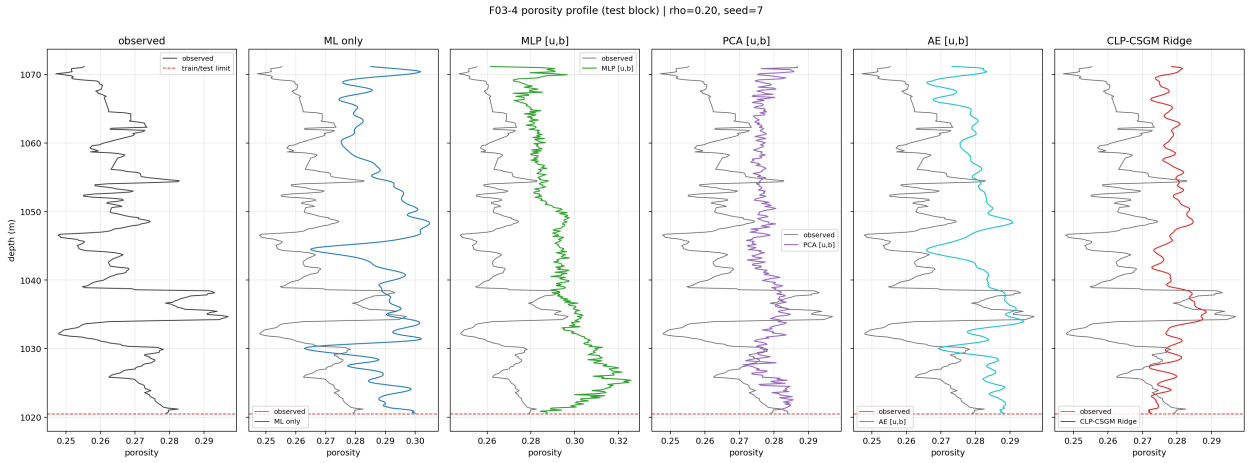

Figure 14: Depth-profile reconstruction for the F03-4 test block at $\rho = 0.20$. CLP-CSGM Ridge follows the observed porosity trend while avoiding the unstable oscillations of direct MLP $[u, b]$.

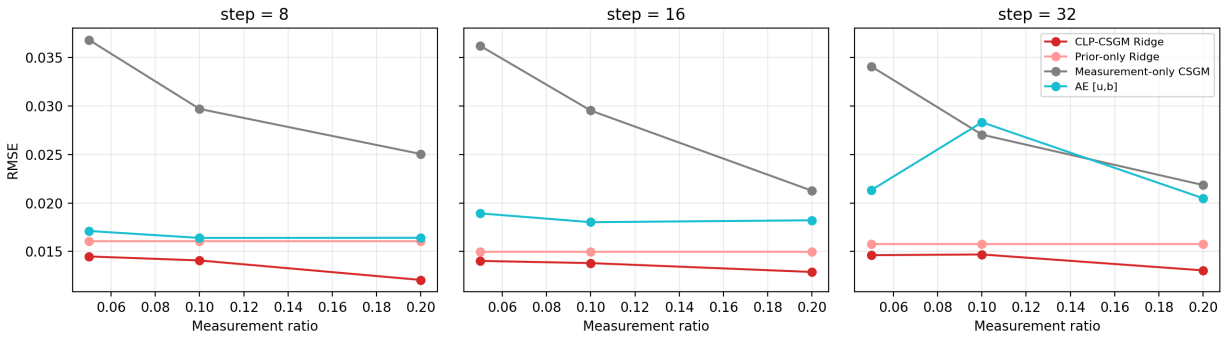

Figure 15: Cross-well Vc ablation RMSE. CLP-CSGM Ridge consistently improves over the prior-only decoder and measurement-only CSGM across the low-data grid.

It also improves over measurement-only CSGM at all ratios, with the largest gains in the sparse regime. The comparison with AE $[u, b]$ remains regime-dependent: CLP-CSGM is better at $\rho = 0.20$ and $\rho = 0.30$, while AE is slightly better at higher ratios.

Table 7: Real-well F03-4 GR-only ablation results. Gains are RMSE reductions of CLP-CSGM Ridge relative to the corresponding ablation.

ρ	CLP-CSGM	Prior-only	Meas.-only	AE $[u, b]$	Gain vs prior (%)	Gain vs meas. (%)
0.2000	0.0158	0.0194	0.0258	0.0180	18.8931	38.9402
0.3000	0.0145	0.0194	0.0228	0.0154	25.3693	36.4859
0.4000	0.0134	0.0194	0.0206	0.0130	30.8973	34.8224
0.5000	0.0125	0.0194	0.0164	0.0123	35.8656	24.1136
0.6000	0.0119	0.0194	0.0150	0.0113	38.9445	21.1069

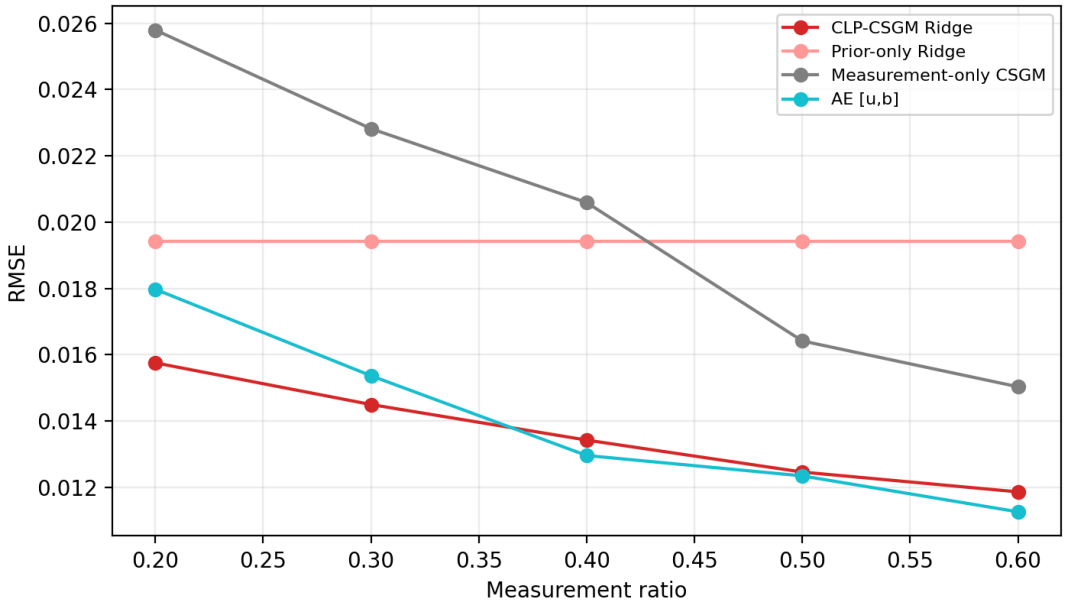

Figure 16: Real-well F03-4 GR-only ablation RMSE. The full method is better than both structural ablations across all measurement ratios, while its comparison with AE $[u, b]$ depends on the sparse-measurement regime.

Finally, we compare the Ridge latent prior against a more flexible MLP prior using the same experimental grids, without repeating the structural ablations. This is a prior-class sensitivity check rather than an architecture search. Table 8 shows that Ridge outperforms the MLP prior in all nine cross-well cells, with RMSE reductions from 8.4% to 33.9%. Table 9 shows the same pattern on F03-4: Ridge is better at every measurement ratio, with the largest gains in the sparse regime. These results support the use of the low-capacity Ridge prior as the paper-facing variant in the studied low-data settings. Figure 17 summarizes this comparison visually: the Ridge curves remain below the MLP curves on both datasets, indicating that the more flexible prior does not improve the latent anchor under the available training data.

Table 8: Cross-well Vc prior-class sensitivity. Ridge gain is the RMSE reduction of CLP-CSGM Ridge relative to CLP-CSGM MLP.

Step	ρ	CLP-CSGM MLP	CLP-CSGM Ridge	Ridge gain (%)
8	0.0500	0.0183	0.0145	20.8838
8	0.1000	0.0171	0.0141	17.8452
8	0.2000	0.0138	0.0121	12.5116
16	0.0500	0.0183	0.0140	23.4853
16	0.1000	0.0175	0.0138	21.3098
16	0.2000	0.0141	0.0129	8.4344
32	0.0500	0.0221	0.0146	33.8867
32	0.1000	0.0212	0.0147	30.7005
32	0.2000	0.0158	0.0131	17.4973

Table 9: Real-well F03-4 GR-only prior-class sensitivity. Ridge gain is the RMSE reduction of CLP-CSGM Ridge relative to CLP-CSGM MLP.

ρ	CLP-CSGM MLP	CLP-CSGM Ridge	Ridge gain (%)
0.2000	0.0186	0.0158	15.3917
0.3000	0.0167	0.0145	13.0313
0.4000	0.0152	0.0134	11.6024
0.5000	0.0132	0.0125	5.3387
0.6000	0.0124	0.0119	4.3866

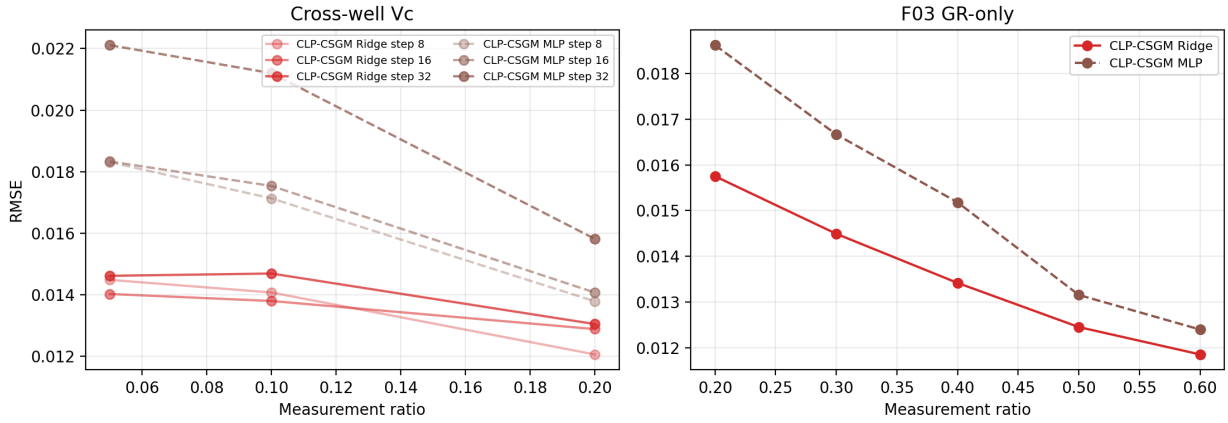

Figure 17: Prior-class sensitivity for CLP-CSGM. The Ridge prior is consistently better than the MLP prior on both the cross-well and F03 GR-only grids.

7 Discussion

The results support the central hypothesis: sparse target-well measurements are more useful when they are enforced through a measurement-consistency term over a learned latent manifold than when they are used only as additional supervised features. In the cross-well Vc benchmark, the training set is small and the held-out well has a shifted target distribution. Direct baselines must learn a complete $[u, b] \to y$ map from few windows. CLP-CSGM Ridge decomposes the task into two simpler components: learn a plausible target manifold from training windows, then select a nearby latent code that is consistent with sparse measurements in the target well.

The ridge prior is important in this decomposition. It provides a stable log-conditioned center $z_0(u)$ in regimes where a more flexible predictor could overfit. The sparse measurements then act as a local correction rather than as the only source of information. This explains why the method performs especially well in low-data and low-ρ settings.

The ablation results strengthen this interpretation. The prior-only decoder is weaker than the full method, which shows that sparse measurements add useful target-well information beyond the log-conditioned latent estimate. The measurement-only CSGM variant is also weaker, especially at low ρ, which shows that the sparse observations alone do not sufficiently identify a stable latent code. The method works best when both sources are used together: logs anchor the latent search, and sparse measurements correct it.

The Ridge-versus-MLP sensitivity study further supports the low-data design choice. A more flexible MLP prior does not improve the conditional latent prior in the tested grids; it is consistently worse than Ridge. This suggests that, with few training windows, regularized linear latent prediction is preferable to a higher-capacity prior that can overfit or become unstable.

The real-well F03 experiment gives a more nuanced result. CLP-CSGM Ridge is the best method on average and is strongest when ρ is small. However, AE $[u, b]$ slightly outperforms it at larger ρ. This loss is not inconsistent with the method objective. CLP-CSGM is designed for sparse assimilation under a generative prior; when many target samples are available, a direct model can exploit them without paying the cost or bias of staying near the learned latent manifold.

Several limitations remain. First, the autoencoder is intentionally small, so representation error may limit reconstruction of unusual geological patterns. Second, the theoretical guarantees of CSGM rely on assumptions such as suitable measurement operators and generator ranges; our coordinate subsampling and field data do not directly satisfy a proven guarantee. Third, latent optimization is nonconvex and more expensive than direct regression. Finally, the experiments use a limited number of wells, so broader geological validation is needed before claiming field-wide generality.

These limitations also clarify the contribution. CLP-CSGM Ridge is not a replacement for all direct petrophysical regressors. It is a targeted method for regimes where dense logs provide a useful but incomplete prior and sparse core measurements should be treated as measurement evidence in the target well.

8 Conclusion

We introduced CLP-CSGM Ridge, a conditional generative compressed sensing method for petrophysical property estimation from dense logs and sparse core measurements. The method learns a target-property decoder, predicts a log-conditioned latent prior with ridge regression, and refines the latent code at test time by enforcing sparse measurement consistency.

In cross-well low-data Vc prediction, CLP-CSGM Ridge outperformed the strongest direct $[u, b]$ autoencoder baseline in all tested training-size and measurement-ratio cells. In the real-well F03

GR-only porosity case, it achieved the best average RMSE, with its clearest advantage in the sparsest measurement regimes. These results indicate that conditional latent generative priors are a promising alternative to fixed-basis sparse residual recovery for petrophysical sparse-measurement assimilation.

The ablation results clarify why the method works in the studied regimes. The prior-only decoder $G(h(u))$ is weaker than the full method, showing that sparse target-well measurements add information beyond log-conditioned latent regression. Measurement-only CSGM is also weaker, showing that sparse observations alone do not provide a stable latent estimate in these low-data settings. Finally, the Ridge prior is consistently stronger than the MLP prior, supporting the use of a low-capacity conditional latent prior when training windows are scarce.

Future work should evaluate the method on additional fields, compare richer conditional priors, investigate uncertainty quantification in latent space, and study measurement designs beyond coordinate subsampling.

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